HewaSense Retrospective Validation

Testing Climate Triggers Against Ten Years of National Yield Data

January 2026

Executive Summary

This document outlines a retrospective validation approach for the HewaSense parametric insurance platform using Tanzania's national rice yield data from 2015 to 2025. The methodology tests whether your machine learning climate forecasts and threshold-based triggers would have successfully predicted the actual crop failures and successes that Tanzanian rice farmers experienced over the past decade.

The approach centers on a critical natural experiment: the catastrophic 2021/2022 season when national yields collapsed to 2.00 metric tons per hectare despite farmers expanding planted area to a record 1.351 million hectares. This represents exactly the kind of systemic climate shock that parametric insurance must detect and trigger on. By testing whether your April rainfall threshold of 120 millimeters would have predicted this disaster nine months in advance, you can validate both the sensitivity of your system to true losses and its specificity against false alarms during good years.

1. Tanzania National Rice Production Dataset (2015–2025)

The following table presents the complete longitudinal record that forms the foundation for retrospective validation. Each marketing year runs from October to September, aligning with Tanzania's rice growing calendar. The flowering phase for rice typically occurs in March through April, which is the critical period when water stress has the most severe impact on yield formation.

Marketing Year	Area (1000 Ha)	Yield (MT/Ha)	Classification
2015/2016	1,154	2.54	Moderate
2016/2017	1,054	3.20	**Good**
2017/2018	1,097	2.24	**Loss**
2018/2019	1,033	3.31	**Good**
2019/2020	1,053	3.30	**Good**
2020/2021	1,100	2.76	Moderate
2021/2022	**1,351**	**2.00**	**CATASTROPHE**
2022/2023	1,000	3.33	**Good**
2023/2024	1,100	3.37	**Good**
2024/2025	1,125	3.39	**Good**

This dataset reveals three distinct patterns that your validation must account for. The 2021/2022 catastrophe represents the ultimate test case, where yields fell to 2.00 metric tons per hectare despite record planted area. The three consecutive strong years from 2022 through 2024 test your system's ability to avoid false positives during favorable conditions. The 2017/2018 decline to 2.24 metric tons per hectare provides an intermediate test case for detecting moderate stress events.

1.1 Kilombero Valley Context

The Kilombero Valley contributes approximately nine percent of national rice production and exhibits systematic yield gaps relative to national averages. Baseline yields under typical smallholder management range from 1.2 to 1.8 metric tons per hectare for local varieties like Supa, compared to the national average of 3.37 metric tons per hectare. This means that when national yields decline by thirty percent, Kilombero farmers experience proportionally larger absolute losses that push them below subsistence thresholds.

Variety	Yield Range (MT/Ha)	Context
SARO 5	2.6 – 4.3	Improved variety; semi-aromatic
Supa (Local)	**1.0 – 1.8**	Most common; farmer preferred
TARI-RIC1	1.5 – 3.5	Morogoro recommendation
GAP Trials	**2.6 – 9.7**	Research conditions; optimal inputs

The critical threshold for Kilombero farmers appears to be approximately 1.0 metric ton per hectare. Below this level, production does not cover variable costs of inputs and labor. When national yields drop to 2.00 metric tons per hectare as in 2021/2022, the proportional calculation suggests Kilombero yields fell to approximately 0.7 to 1.0 metric tons per hectare, representing true catastrophic loss. This provides the empirical basis for defining when your trigger system should activate.

2. Retrospective Validation Methodology

The validation approach tests whether your machine learning climate forecasts, combined with the 120 millimeter April rainfall threshold, would have correctly identified the years when farmers experienced actual losses. This is fundamentally a binary classification problem with four possible outcomes for each year.

2.1 The Four Validation Outcomes

True Positive (Correct Trigger): Your system predicted April rainfall below 120 millimeters, and the subsequent harvest showed yields below the loss threshold. This represents the ideal outcome where insurance pays farmers who actually experienced losses. The 2021/2022 season is the critical test case that must produce a true positive for your system to be considered valid.

True Negative (Correct Non-Trigger): Your system predicted April rainfall above 120 millimeters, and the harvest was successful with yields above the loss threshold. This demonstrates that your system correctly identified favorable conditions and avoided unnecessary payouts. The 2018/2019, 2019/2020, 2022/2023, 2023/2024, and 2024/2025 seasons should ideally produce true negatives.

False Negative (Missed Loss): Your system predicted April rainfall above 120 millimeters, but the harvest failed with yields below the loss threshold. This is the worst possible outcome for farmer welfare, as it represents uninsured losses. Farmers who paid premiums receive no payout despite experiencing crop failure. This is basis risk in its most damaging form. If the 2021/2022 collapse produces a false negative, your threshold calibration requires immediate revision.

False Positive (Unnecessary Payout): Your system predicted April rainfall below 120 millimeters and triggered a payout, but the harvest was successful with yields above the loss threshold. While less harmful than false negatives, false positives deplete capital reserves and force premium increases in subsequent years. Too many false positives make the product unsustainable. If any of the strong yield years from 2022 through 2024 produce false positives, this suggests your threshold may be too conservative.

2.2 Defining the Loss Threshold

To classify each year as a true loss year or successful year, you need to establish a yield threshold below which farmers experience economically significant damage. Based on the Kilombero context, you can define multiple threshold levels to test different insurance product designs.

Threshold	Loss Years	Product Type	Premium Implications
< 2.00 MT/Ha	2021/2022 only	Catastrophic-only	Very low premiums
< 2.50 MT/Ha	**2017/2018, 2021/2022**	**Balanced coverage**	Moderate premiums
< 2.60 MT/Ha	Add 2015/2016, 2020/2021	Moderate stress	Higher premiums

The choice of threshold reflects your product design philosophy. A catastrophic-only product would use the 2.00 metric ton per hectare threshold, triggering only in the worst years like 2021/2022. A moderate stress product would use 2.60 metric tons per hectare, also covering years like 2020/2021. For the initial validation, testing against the 2.50 metric ton per hectare threshold provides a balanced approach that captures true catastrophes while filtering out normal yield variation.

3. Step-by-Step Implementation Guide

This section provides concrete code examples and systematic procedures for running the retrospective validation. The process involves three main steps: querying historical climate forecasts, applying the trigger logic, and calculating performance metrics.

3.1 Step One: Query Historical Climate Data

For each marketing year from 2015/2016 through 2024/2025, you need to retrieve the April rainfall forecast that your machine learning model would have generated. If you have historical forecasts stored in a database, this is straightforward retrieval. If not, you will run your trained model in hindcast mode, using only the data that would have been available at the time of the forecast.

Critical consideration for hindcast mode: To avoid data leakage, your model must only use climate data from October through March to predict April rainfall. It cannot use data from April or later months, as that would represent information from the future. This maintains the realistic constraint that forecasts must be generated before the flowering phase begins.

```
# Query historical April rainfall forecasts
# for Kilombero Valley rice growing season

import pandas as pd
import numpy as np
from scipy.stats import norm

# Define the marketing years to test
marketing_years = [
    '2015/2016', '2016/2017', '2017/2018',
    '2018/2019', '2019/2020', '2020/2021',
    '2021/2022', '2022/2023', '2023/2024',
    '2024/2025'
]

# For each year, query the forecast that would have been
# generated in early April (before flowering)
results = []

for year in marketing_years:
    # Extract the year when April occurs
    # e.g., 2015/2016 season → April 2016
```

```
    april_year = int(year.split('/')[1])
    forecast_date = f'{april_year}-04-01'

    # Query your ML model or database
    # This returns: predicted_rainfall, confidence, std_dev
    forecast = query_climate_model(
        location='Kilombero',
        date=forecast_date,
        variable='monthly_rainfall'
    )

    results.append({
        'marketing_year': year,
        'april_year': april_year,
        'predicted_rainfall': forecast['mean'],
        'forecast_confidence': forecast['confidence'],
        'std_deviation': forecast['std']
    })
```

3.2 Step Two: Calculate Event Probabilities

Now that you have point predictions for each year, you need to calculate the probability that rainfall will fall below your 120 millimeter threshold. This is distinct from forecast confidence. High confidence in a prediction of 130 millimeters does not mean high risk, because 130 millimeters is safely above the threshold. Conversely, lower confidence in a prediction of 90 millimeters still represents high risk, because even with uncertainty the rainfall is likely to be insufficient.

```
# Calculate probability of adverse event
# P(rainfall < 120mm) for each forecast

THRESHOLD = 120  # mm

for result in results:
    mean = result['predicted_rainfall']
    std = result['std_deviation']

    # Calculate cumulative probability
    # that rainfall < threshold
```

```
    prob_adverse = norm.cdf(THRESHOLD, loc=mean, scale=std)

    result['event_probability'] = prob_adverse

    # Decision logic for trigger
    # Layer 2: Deterministic parametric rule
    result['trigger_activated'] = (mean < THRESHOLD)

    # Risk management flags (for portfolio reserves)
    result['high_risk'] = (prob_adverse >= 0.75)
    result['alert'] = (prob_adverse >= 0.40)
```

3.3 Step Three: Match Triggers to Actual Outcomes

The final step merges your forecast triggers with the actual observed yields to determine whether each prediction was correct. This creates the confusion matrix that quantifies basis risk.

```
# Actual yields from national dataset
actual_yields = {
    '2015/2016': 2.54,
    '2016/2017': 3.20,
    '2017/2018': 2.24,
    '2018/2019': 3.31,
    '2019/2020': 3.30,
    '2020/2021': 2.76,
    '2021/2022': 2.00,
    '2022/2023': 3.33,
    '2023/2024': 3.37,
    '2024/2025': 3.39
}

# Define loss threshold (multiple options to test)
LOSS_THRESHOLD = 2.50  # MT/Ha

# Classify each year
for result in results:
    year = result['marketing_year']
    actual_yield = actual_yields[year]
```

```
    # Did farmers actually experience losses?
    actual_loss = (actual_yield < LOSS_THRESHOLD)

    # Did our trigger activate?
    trigger = result['trigger_activated']

    # Classify outcome
    if trigger and actual_loss:
        result['outcome'] = 'TRUE_POSITIVE'
    elif not trigger and not actual_loss:
        result['outcome'] = 'TRUE_NEGATIVE'
    elif not trigger and actual_loss:
        result['outcome'] = 'FALSE_NEGATIVE'  # Missed loss
    else:  # trigger and not actual_loss
        result['outcome'] = 'FALSE_POSITIVE'  # Unnecessary payout
```

3.4 Step Four: Calculate Performance Metrics

From the classified outcomes, you can now calculate the key metrics that quantify your system's performance. These metrics will be central to your BimaLab application.

```
# Calculate performance metrics
df = pd.DataFrame(results)

true_positives = sum(df['outcome'] == 'TRUE_POSITIVE')
true_negatives = sum(df['outcome'] == 'TRUE_NEGATIVE')
false_positives = sum(df['outcome'] == 'FALSE_POSITIVE')
false_negatives = sum(df['outcome'] == 'FALSE_NEGATIVE')

total = len(df)

# Basis risk: proportion of incorrect classifications
basis_risk = (false_positives + false_negatives) / total

# False negative rate: missed losses (most harmful)
total_loss_years = true_positives + false_negatives
false_negative_rate = false_negatives / total_loss_years
```

```
# False positive rate: unnecessary payouts
total_success_years = true_negatives + false_positives
false_positive_rate = false_positives / total_success_years

# Overall accuracy
accuracy = (true_positives + true_negatives) / total

print(f'Basis Risk: {basis_risk:.1%}')
print(f'False Negative Rate: {false_negative_rate:.1%}')
print(f'False Positive Rate: {false_positive_rate:.1%}')
print(f'Overall Accuracy: {accuracy:.1%}')
```

4. Expected Results and Interpretation Framework

This section helps you understand what the validation results will mean when you receive them. The interpretation depends critically on whether your system correctly predicted the 2021/2022 catastrophe and avoided false triggers during the 2022 through 2024 recovery period.

4.1 Best Case Scenario

If your retrospective validation achieves the following pattern of results, your system demonstrates exceptional performance and will be a compelling candidate for BimaLab support.

Year	Actual Yield	Trigger?	Outcome
2017/2018	2.24	Yes	**True Positive ✓**
2018/2019	3.31	No	**True Negative ✓**
2019/2020	3.30	No	**True Negative ✓**
2020/2021	2.76	Yes	*False Positive (borderline)*
2021/2022	**2.00**	**Yes**	**True Positive ✓✓**
2022/2023	3.33	No	**True Negative ✓**
2023/2024	3.37	No	**True Negative ✓**
2024/2025	3.39	No	**True Negative ✓**
Result	**8/10 correct**	**Basis Risk: 20%**	**EXCELLENT**

Interpreting this outcome: With eight out of ten years classified correctly, your basis risk would be twenty percent, which significantly outperforms the industry baseline of thirty-five to forty percent. Most importantly, you successfully predicted both major loss events while avoiding false alarms during the strong recovery years. The single false positive in 2020/2021 is acceptable, as that year did show moderate yield stress and borderline conditions where a conservative trigger is defensible.

BimaLab application strength: This result provides compelling evidence that your machine learning enhanced parametric system adds genuine value over simple historical thresholds. You can quantify the improvement: basis risk of twenty percent versus industry baseline of thirty-five percent represents a forty-three percent reduction in misclassification error. This translates directly to better farmer outcomes and more sustainable capital reserves.

4.2 Acceptable Scenario

An acceptable but less impressive outcome would correctly identify the 2021/2022 catastrophe but generate one or two false positives during good years, or miss the 2017/2018 warning signal while catching the major event.

Year	Actual Yield	Trigger?	Outcome
2017/2018	2.24	No	False Negative ✗
2019/2020	3.30	Yes	False Positive

2021/2022	**2.00**	**Yes**	**True Positive ✓✓**
2022/2023	3.33	Yes	False Positive
Result	**7/10 correct**	**Basis Risk: 30%**	**ACCEPTABLE**

Interpreting this outcome: With seven out of ten years classified correctly, your basis risk is thirty percent, which meets the industry standard but does not significantly exceed it. The critical test of 2021/2022 was passed, demonstrating that your system can detect true catastrophes. However, the two false positives in 2019/2020 and 2022/2023 suggest your threshold may be slightly too conservative or that your uncertainty quantification needs refinement.

BimaLab application strength: This result is still viable for BimaLab but requires additional narrative. You would emphasize that your system prioritizes sensitivity to true losses over minimizing false positives, which is a defensible product design choice for a smallholder protection product. You would also commit to threshold recalibration during the BimaLab program using ground-truth data from the actual deployment to reduce false positive rates.

4.3 Problematic Scenario

A problematic outcome would fail to detect the 2021/2022 catastrophe, producing a false negative. This would be a fundamental failure of the system's core value proposition.

Year	Actual Yield	Trigger?	Outcome
2021/2022	**2.00**	**NO**	**FALSE NEGATIVE ✗✗**
Critical Test	**FAILED**	**Basis Risk: >40%**	**NEEDS REVISION**

Interpreting this outcome: If your system predicted April 2022 rainfall above 120 millimeters when yields subsequently collapsed to 2.00 metric tons per hectare nationally, this indicates that either your machine learning forecasts failed to detect the adverse conditions or your 120 millimeter threshold is fundamentally miscalibrated. This scenario reveals basis risk of forty percent or higher, which is worse than simple historical baselines and suggests the system needs major revision before deployment.

Response if this occurs: Do not apply to BimaLab with this result. Instead, conduct a deep diagnostic to understand what went wrong. Possible causes include your machine learning model's training data not capturing the type of climatic pattern that led to the 2021/2022 failure, the 120 millimeter threshold being too low and needing adjustment to 140 or 150 millimeters, or April rainfall alone being insufficient such that you need to incorporate additional variables like total seasonal rainfall or temperature stress during critical growth stages. Address these issues, revalidate, and apply to BimaLab in a subsequent cohort.

5. Deep Dive: What Happened in 2021/2022?

The 2021/2022 marketing year represents the most important test case for your validation. Understanding what actually happened during this season will help you interpret your results and, if necessary, diagnose why your system did or did not predict the collapse. This section prepares you to conduct a post-hoc analysis once you have the forecast data.

5.1 What We Know From National Aggregates

National yields fell to 2.00 metric tons per hectare, a forty percent decline from the previous year's 3.30 metric tons per hectare. This magnitude of drop is unprecedented in the dataset except for the 2017/2018 event. Critically, farmers expanded planted area to a record 1.351 million hectares, an increase of more than two hundred thousand hectares over the previous year. This behavior pattern reveals that farmers anticipated difficulties and attempted to compensate by planting more area, but the climatic conditions were severe enough that even this maximum effort resulted in catastrophic aggregate losses.

This farmer response is precisely the scenario where parametric insurance provides maximum value. Insurance based on actual yield would observe the low harvest and pay, but with a significant time lag while harvest data is collected and verified. Parametric insurance based on your April rainfall forecast could have provided early warning in March or early April and paid out in May, giving farmers capital to purchase inputs for the following season or covering immediate household needs. The nine-month lead time from trigger to harvest is the critical advantage you need to emphasize.

5.2 Diagnostic Questions for April 2022

When you retrieve your machine learning model's predictions for April 2022, ask yourself these specific questions to understand whether your system worked as designed.

Question One: What did your ensemble models predict for April 2022 total rainfall? You need the individual predictions from Random Forest, XGBoost, and LSTM models, not just the ensemble mean. If the models strongly disagree, this indicates high uncertainty, which should affect your event probability calculation.

Question Two: Was the predicted rainfall above or below your 120 millimeter threshold? If your mean prediction was 115 millimeters, your system would have triggered correctly. If it was 135 millimeters, your system missed the event. The margin matters: a prediction of 122 millimeters that barely missed the trigger is very different from a prediction of 180 millimeters that failed to anticipate stress.

Question Three: What was the actual observed April 2022 rainfall in Kilombero? Retrieve ground station data or satellite estimates for validation. If actual rainfall was 85 millimeters, your forecast should have been in that range. If actual rainfall was 140

millimeters, the crop failure was not due to April water stress, and your threshold logic needs revision to account for stress in other months or growth stages.

Question Four: What does the full seasonal rainfall pattern reveal? Look at cumulative rainfall from November 2021 through May 2022. Rice is sensitive to water stress throughout the season, not just during flowering. If the 2021/2022 failure was driven by early-season drought in November through January that delayed planting and reduced crop establishment, your April-only trigger would not capture this. You may need to add triggers for multiple growth phases.

5.3 Potential Explanations for Different Outcomes

Scenario A: Your system triggered correctly (True Positive). If your April 2022 prediction was below 120 millimeters and actual rainfall was also low, this validates both your machine learning forecasting capability and your threshold calibration. Your BimaLab narrative becomes: Our system correctly identified the most severe rice production crisis in Tanzania's recent history nine months before the harvest failure became apparent.

Scenario B: Your system almost triggered but barely missed. If your April 2022 prediction was 125 millimeters with high uncertainty, such as a standard deviation of 20 millimeters, your event probability P(rain < 120mm) would have been around forty percent, flagging an alert but not a high-risk trigger. This is a borderline case that suggests your threshold might need minor adjustment to 130 millimeters. Your BimaLab narrative emphasizes that fine-tuning thresholds is exactly what the shadow deployment will accomplish.

Scenario C: Your system completely missed the event. If your April 2022 prediction was 160 millimeters and actual observed rainfall was also high, the crop failure was not driven by flowering-phase water stress. This requires investigating alternative explanations. Were there floods rather than droughts? Did excessive rainfall during grain-filling cause disease outbreaks? Was there a temperature shock or pest epidemic? This scenario means your single-phase, rainfall-only trigger is incomplete and needs expansion to multi-variable or multi-phase logic.

6. Recommendations and Next Steps

Based on this validation framework, here are concrete recommendations for completing the analysis.

6.1 Immediate Action Items (This Week)

Run the retrospective validation code exactly as provided in Section Three. Query your machine learning model for all ten marketing years and generate the complete results table with predicted rainfall, event probabilities, trigger decisions, and outcome classifications. This will take your model's current configuration as given and test whether it performs adequately without any modifications.

Calculate the four key metrics: Basis risk, false negative rate, false positive rate, and overall accuracy. Compare these to the target thresholds in your original specifications. Basis risk should be below twenty-five percent, false negative rate below fifteen percent, and false positive rate below twenty percent to claim superior performance relative to industry baselines.

Share the complete results table for deep analysis. Once you have generated the results, provide the full table showing each year's predictions, actual yields, trigger decisions, and outcome classifications. This will enable a detailed diagnostic to understand which years were classified correctly and which were not, and what adjustments might improve performance.

6.2 Depending on Results: Calibration or Application

If results meet targets (basis risk below twenty-five percent): Proceed directly to BimaLab application preparation. Your validation demonstrates that the system works with the current 120 millimeter threshold, and no fundamental changes are needed. Compile the results into a white paper showing the year-by-year predictions, the confusion matrix, and the quantified basis risk reduction versus industry baselines. Emphasize the successful prediction of 2021/2022 catastrophe and clean performance during the 2022 through 2024 recovery.

If results are borderline (basis risk between twenty-five and thirty-five percent): Conduct threshold optimization. Test alternative thresholds from 100 millimeters to 150 millimeters in 10 millimeter increments to find the value that minimizes basis risk. This is a straightforward parameter sweep that requires rerunning the validation code with different threshold values. Document the optimization process and show that your final threshold selection is data-driven. Mention in the BimaLab application that threshold refinement through shadow deployment is a planned activity.

6.3 Key Components

The following evidence-based components demonstrate the system's viability.

Component One: Ten-Year Retrospective Validation Results. Present the complete table showing predictions and outcomes for each year from 2015 to 2025. Highlight the 2021/2022 test case as the centerpiece of your validation.

Component Two: Quantified Basis Risk Comparison. Create a visual comparison showing your achieved basis risk versus the industry baseline of thirty-five to forty percent. Express the improvement as a percentage reduction in misclassification error.

Component Three: Technical Architecture Documentation. Show the two-layer architecture with Layer One doing machine learning climate forecasting and Layer Two applying deterministic threshold logic. Explain why this separation is critical for regulatory compliance, farmer trust, and fail-safe operation.

Component Four: Shadow Deployment Plan. Propose a two-year deployment with five hundred to one thousand farmers in Kilombero where you issue trigger decisions alongside traditional yield assessments. This will generate ground-truth data on basis risk in real operational conditions and allow refinement of thresholds before scaling to full commercial deployment.

7. Conclusion: From Validation to Deployment

The retrospective validation approach outlined in this document provides a rigorous test of whether your HewaSense platform delivers on its core value proposition: reducing basis risk in agricultural parametric insurance through machine learning climate forecasts. By testing your system against ten years of actual national yield outcomes, you move beyond theoretical claims about forecast accuracy to empirical evidence of insurance product performance.

The 2021/2022 marketing year serves as the ultimate test case. This season saw national rice yields collapse to 2.00 metric tons per hectare despite farmers expanding cultivated area to record levels, indicating a systemic climate shock that overwhelmed farmer adaptation efforts. If your April rainfall forecast successfully predicted this event nine months before harvest, you can credibly claim that your system provides the early-warning capability that makes parametric insurance valuable for smallholder risk management.

Equally important are the three consecutive strong yield years from 2022 to 2024. Your system must avoid false triggers during these periods to demonstrate specificity. False positives that pay farmers who did not experience losses are sustainable only if they are rare exceptions, not systematic errors. The validation results will quantify both sensitivity and specificity, giving you the complete picture of product performance.

Once you complete this validation and share the results, we can conduct a deep diagnostic analysis to understand what drove the outcomes in each year and whether any calibration adjustments are needed before BimaLab application. The framework is ready. Now execute the code, retrieve the predictions, and let the data reveal whether your system meets the twenty-five percent basis risk target that would position HewaSense as a genuine advance over existing parametric insurance approaches.

Document prepared: January 2026

Ready for immediate implementation